P 2 Pharmaceutical Development

Enclosed Herewith

PRODUCT DEVELOPMENT REPORT

SOLIFENACIN SUCCINATE TABLETS 5 MG

PRODUCT DEVELOPMENT REPORT

TABLE OF CONTENTS

1.1	Information on Development Studies	
1.2	Components of Drug Product	
	1.2.1	Active Substance
	1.2.2	Excipients
1.3	Finished Product	
	1.3.1	Formulation Development
	1.3.2	Overages
	1.3.3	Physicochemical and Biological Properties
1.4	Manufacturing Process Development	
1.5	Container Closure System	
1.6	Microbiological Attributes	
1.7	Compatibility	

1.1 INFORMATION ON DEVELOPMENT STUDIES

The Pharmaceutical developmental studies have to be carried out with the purpose of selecting the right dosage form and a stable formulation. These studies give detailed description of all the steps involved in the process of development of the finished product. Such details are in tended towards identifying the critical parameters involved in the developmental process, which have to be controlled in order to give reliable and reproducible quality product.

Rational for Dosage Form:

Each film coated tablet contains:

Solifenacin Succinate BP.....5 mg

Excipients.....q.s.

The Development report has been prepared to elaborate following:

- Components of the Product
- Formulation development
- Physiochemical and Biological Properties
- Details of Developmental Stages of formulated product
- Manufacturing Process development
- Critical Process Parameters
- Microbiological Attributes
- Compatibility
- Conclusion

LEBEN LIFE SCIENCES	**Leben Life Sciences Pvt. Ltd.** Akola – 444104 (MS) INDIA
Title	**PRODUCT DEVELOPMENT REPORT**
Product	**SOLIFENACIN SUCCINATE TABLETS 5 MG**

TARGET PRODUCT ATTRIBUTE PROFILE:

Sr. No.	Tests	Specifications
1.	**Description**	Light yellow coloured, round shaped, film coated tablet, plain on both sides.
2.	**Identification By IR**	The infrared absorption spectrum of the dried residue is concordant with the reference spectrum of Solifenacin Succinate RS/WS.
3.	**Average weight**	153.00 mg ± 7.5% (141.53 mg to 164.48 mg)
4.	**Uniformity of weight**	Not More than 2 tablets deviate from average weight by 7.5% None Tablet deviate from the average weight by more than 15%
5.	**Content Uniformity**	AV NMT 15
6.	**Disintegration Time**	NMT 30 minutes
7.	**Dissolution**	NLT 75% (Q) of the labeled amount
8.	**Assay**	95.0% to 105.0% of the stated amount
9.	**Microbial Limit Test**	TAMC: NMT 1000 cfu/gm. TYMC: NMT 100 cfu/gm. Escherichia coli: absent/gm. Salmonella: absent/10 gm. P. aeruginosa: absent/gm. S. aureus : absent/gm.

1.2 COMPONENTS OF DRUG PRODUCT

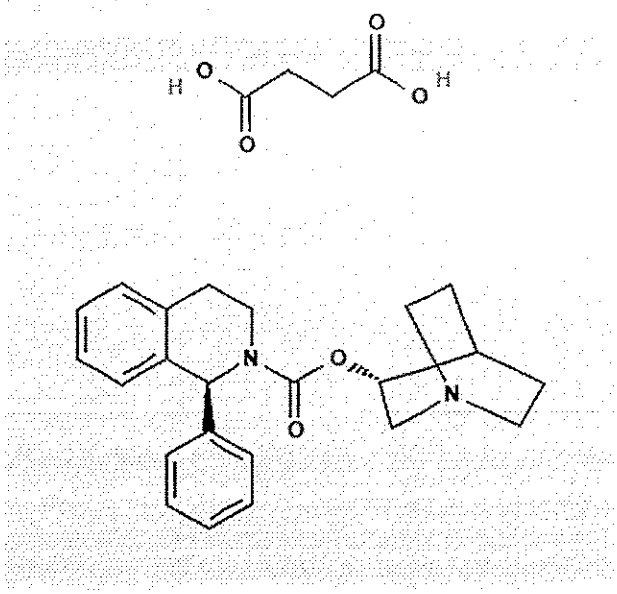
1.2.1 ACTIVE INGREDIENT

Name: Solifenacin Succinate

PREFORMULATION STUDY &SELECTION OFEXCIPIENTS:

Physicochemical properties of Active drug:-

Structure:



Molecular Formula: $C_{27}H_{32}N_2O_6$

Molecular Weight: 480.6 gm/mol

Physical Description: White or light yellow powder.

Solubility Characteristic: Very soluble or freely soluble in water, soluble in ethanol (96 per cent), practically insoluble in heptane.

Mode of action:

Solifenacin is a competitive muscarinic receptor antagonist. It has the highest affinity for M3, M1, and M2 muscarinic receptors. 80% of the muscarinic receptors in the bladder are M2, while 20% are M3. Solifenacin's antagonism of the M3 receptor prevents contraction of the detrusor muscle, while antagonism of the M2 receptor may prevent contraction of smooth muscle in the bladder.

Pharmacokinetic properties

Absorption

Solifenacin is well absorbed in the duodenum, jejunum, and ileum but not the stomach. Absorption occurs via passive diffusion and so no transporters are involved. The mean oral bioavailability of solifenacin is 88%. The T_{max} of solifenacin is 3-8 hours with a C_{ss} of 32.3ng/mL for a 5mg oral dose and 62.9ng/mL for a 10mg oral dose.

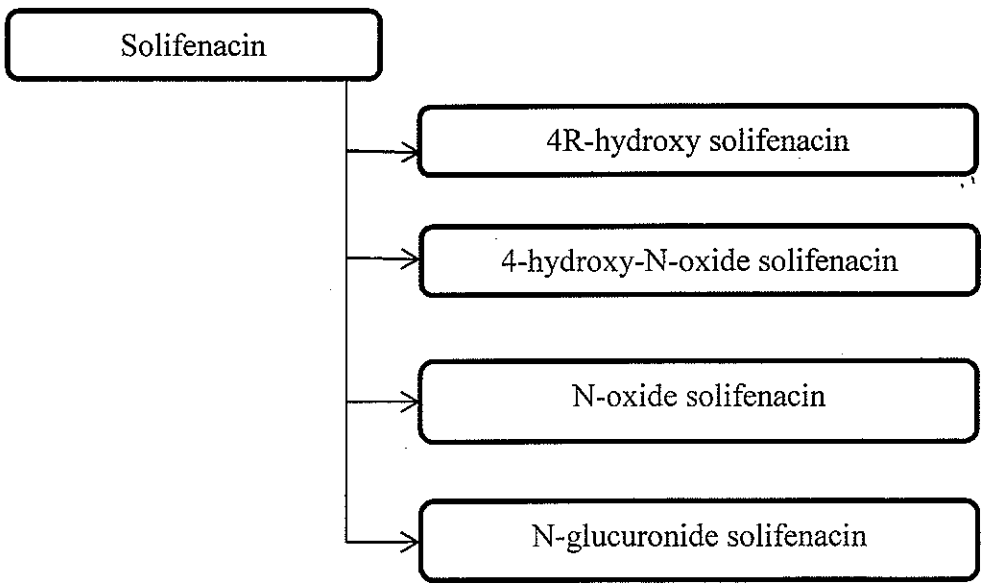
Distribution:

The volume of distribution of solifenacin is 600L.

Metabolism:

Solifenacin undergoes N-oxidation at the quinuclidin ring by cytochrome P450, though the exact enzymes are not revealed in the literature. The tetrahydroisoquinolone ring is 4R-hydroxylated by CYP3A4, CYP1A1, and CYP2D6. A 4R-hydroxy N-oxide metabolite is also formed by CYP3A4. Finally, solifenacin can undergo direct glucuronidation. Only solifenacin and the 4R-hydroxy metabolite are pharmacologically active.

Hover over products below to view reaction partners



Elimination:

69.2±7.8% of a radiolabelled dose is recovered in the urine, 22.5±3.3% was recovered in feces, and 0.4±7.8% was recovered in exhaled air. 18% of solifenacin is eliminated as the N-oxide metabolite, 9% is eliminated as the 4R-hydroxy N-oxide metabolite, and 8% is eliminated as the 4R-hydroxy metabolite.

PREFORMULATION STUDY

Investigation of Physico-chemical properties of our drug component that will affect drug performance and development of our dosage for is described below:

Goal: The rational development of a dosage form of a drug substance alone and when combined with excipients.

Objective: To generate useful information to the formulation or to design an optimum drug delivery system.

Stage 1: Active sourcing

1) The supplier of drug substance is Amoli Organics Pvt. Ltd.
2) Active drug substance is complying with British Pharmacopoeia.
3) Innovator: Vesicare 5 mg (Solifenacin Succinate Tablets 5 mg), Manufactured by Astellas Pharma Europe B.V. B. No : 21E1009

Stage 2: Innovator Product Profile:

Sr. No.	Test	Specification	RLD (Batch No. 21E1009)
1.	Description	Yellow colour, round shaped, film coated tablet, embossing, imprinting on one side and plain on other side.	Yellow colour, round shaped, film coated tablet, embossing, imprinting on one side and plain on other side.
2.	Identification by IR	The infrared absorption spectrum of the dried residue is concordant with the reference spectrum of Solifenacin Succinate RS/WS.	Complies

3.	Content Uniformity	AV NMT 15	1.45
4.	Dissolution	NLT 75 % (Q)	93.68%
5.	Assay	95.0 % to 105.0% of the stated amount.	99.5%

1.2.2 EXCIPIENTS

Based on literature study and desired characteristics of the drug product, the excipients selected for the formulation development of Solifenacin Succinate Tablets 5 mg are tabulated below along with reported concentration, level usage and reason for usage:

Sr. No	Name of Excipients	Pharmacopoeial reference	Function	Level of usage (% w/w)
1	Maize Starch	BP	Bulking Agent	26.14
2	Lactose Monohydrate	BP	Diluent	68.33
3	Hypromellose	BP	Binder	0.67
4	Magnesium Stearate	BP	Lubricant	1.00
5	Ready Coat Iron Oxide Yellow 36F0193220	IH	Film Former	1.96
6	Purified water	BP	Solvent	q.s.

δ= Handbook of Pharmaceutical excipients Sixth Edition.

Excipient Grade Selection:

Maize Starch, Lactose Monohydrate, Hypromellose, Magnesium Stearate and Purified Water was added to impart the required characteristics to the drug product and use of same in developed Solifenacin Succinate Tablets 5 mg is not foreseen to impact the product quality as these are most widely used excipients. Current stability studies of the exhibit batches further support the use of Maize Starch, Lactose Monohydrate, Hypromellose, Magnesium Stearate and Purified water. The selection of excipients grade was based on literature knowledge about excipients that have been used widely in approved products. The selected grade of excipients and level to be used in the formulation was further verified during pharmaceutical development.

DETAILS OF EACH EXCIPIENT:

1. Maize Starch

Description

Maize Starch occurs as an odorless and tasteless, fine, white to off-white powder. It consists of very small spherical or ovoid granules or grains whose size and shape are characteristic for each botanical variety

Functional Category

Diluent, Disintegrate, Binder and Thickening agent.

Applications in Pharmaceutical Formulation or Technology

Maize starch is a versatile excipient used primarily in oral solid-dosage formulations where it is utilized as a binder, diluent, and disintegrant. As a diluent, starch is used for the preparation of standardized triturates of colorants, potent drugs, and herbal extracts, facilitating subsequent mixing or blending processes in manufacturing operations. Maize starch is also used in dry-filled capsule formulations for volume adjustment of the fill matrix, and to improve powder flow, especially when using dried starches. Maize starch quantities of 3–10% w/w can act as an antiadherent and lubricant in tableting and capsule filling. In tablet formulations, freshly prepared starch paste is used at a concentration of 3–20% w/w (usually 5–10%, depending on the starch type) as a binder for wet granulation. The required binder ratio should be determined by optimization studies, using parameters such as tablet friability and hardness, disintegration time, and drug dissolution rate.

2. Lactose Monohydrate

Description

In the solid state, lactose appears as various isomeric forms, depending on the crystallization and drying conditions, i.e. a lactose monohydrate, b-lactose anhydrous, and a-lactose anhydrous. The stable crystalline forms of lactose are a-lactose monohydrate, b-lactose anhydrous and stable a-lactose anhydrous. Lactose occurs as white to off-white crystalline particles or powder. Lactose is odorless and slightly sweet-tasting; a-lactose is approximately 20% as sweet as sucrose, while b-lactose is 40% as sweet.

Functional Category

Diluent

Applications in Pharmaceutical Formulation or Technology

Lactose is widely used as a filler and diluent in tablets and capsules, and to a more limited extent in lyophilized products and infant formulas. (1–9) Lactose is also used as a diluent in dry-powder inhalation; see Lactose, Inhalation. Various lactose grades are commercially available that have different physical properties such as particle size distribution and flow characteristics. This permits the selection of the most suitable material for a particular application; for example, the particle size range selected for capsules is often dependent on the type of encapsulating machine used.

3. Hypromellose

Description

Hypromellose is an odorless and tasteless, white or creamy-white fibrous or granular powder

Functional Category

Binder

Applications in Pharmaceutical Formulation or Technology

Hypromellose is widely used in oral, ophthalmic, nasal, and topical pharmaceutical formulations. In oral products, hypromellose is primarily used as a tablet binder,(1) in film-coating,(2–7) and as a matrix for use in extendedrelease tablet formulations.(8–12) Concentrations between 2% and 5% w/w may be used as a binder in either wet- or dry-granulation processes. High-viscosity grades may be used to retard the release of drugs from a matrix at levels of 10–80% w/w in tablets and capsules. Hypromellose is also used in liquid oral dosage forms as a suspending and/or thickening agent at concentrations ranging from 0.25–5.0%.(13) Depending upon the viscosity grade, concentrations of 2–20% w/w are used for film-forming solutions to film-coat tablets. Lowerviscosity grades are used in aqueous film-coating solutions, while H 326 Hypromellose higher-viscosity grades are used with organic solvents. Examples of film-coating materials that are commercially available include Any Coat C, Spectracel, Pharmacoat, and the Methocel E Premium LV series. Hypromellose is also used as a suspending and thickening agent in topical formulations. Compared with methylcellulose, hypromellose produces aqueous solutions of greater clarity, with fewer undissolved fibers present, and is therefore preferred in formulations for ophthalmic use. Hypromellose at concentrations between 0.45–1.0% w/w may be added as a thickening agent to vehicles for eye drops and artificial tear solutions. It is also used commercially in liquid nasal formulations at a concentration of 0.1 %. (13) Hypromellose is used as an emulsifier, suspending agent, and stabilizing agent in topical gels and

ointments. As a protective colloid, it can prevent droplets and particles from coalescing or agglomerating, thus inhibiting the formation of sediments. In addition, hypromellose is used in the manufacture of capsules, as an adhesive in plastic bandages, and as a wetting agent for hard contact lenses. It is also widely used in cosmetics and food products.

4. Magnesium stearate

Description

Magnesium stearate is a very fine, light white, precipitated or milled, impalpable powder of low bulk density, having a faint odor of stearic acid and a characteristic taste. The powder is greasy to the touch and readily adheres to the skin.

Functional Category

Lubricant

Applications in Pharmaceutical Formulation or Technology

Magnesium stearate is widely used in cosmetics, foods, and pharmaceutical formulations. It is primarily used as a lubricant in capsule and tablet manufacture at concentrations between 0.25% and 5.0% w/w. It is also used in barrier creams.

Selection of Excipients:

The excipients used in Solifenacin Succinate Tablets 5 mg were selected based on the literature search and excipients used in Reference Listed Drug.

The summary of excipient-drug substance compatibility studies provided below and the selection of each excipient grade is provided in the following Section

S. No	Name of Excipients	Pharmacopoeial reference	Function
1	Maize Starch	BP	Bulking Agent
2	Lactose Monohydrate	BP	Diluent
3	Hypromellose	BP	Binder
4	Magnesium stearate	BP	Lubricant
5	Ready Coat Iron Oxide Yellow 36F0193220	IH	Film Former
6	Purified water	BP	Solvent

Drug-Excipient Compatibility study

All ingredients used, are well known and widely used as pharmaceutical excipients and comply with the relevant Pharmacopoeial monographs. The choice of excipients meets the requirements of satisfactory pharmaceutical properties required for development of Oral solid dosage form. The quantities of the excipients were selected from the recommended ranges stated in the literature (Handbook of Pharmaceutical Excipients Edition: 6) and were further optimized in the course of various lab trials conducted during formulation development.

Drug-Excipient Compatibility study Table:

Sr No.	Drug and Excipients Mixture	Drug: Excipient ratio	Condition	
			1)40°C/75%RH (Open) 2)25°C/60%RH (Open) 30 Days	1) 40°C/75%RH (Closed) 2) 25°C/60%RH (Closed) 30 Days
1	Solifinacine Succinate	100 mg	✓	✓
2	API+ Lactose Monohydrate	1 : 1	✓	✓
4	API + Maize starch	1 : 1	✓	✓
6	API + Hypromellose	1 : 1	✓	✓
7	API + Magnesium Stearate	1 : 1	✓	✓
8	API + Purified Water	1 : 1	✓	✓
9	API + Ready Coat Iron Oxide Yellow 36F0193220	1 : 1	✓	✓
10	API + All Excipeints	1 : 1	✓	✓

Placebo Details:			
Lactose Monohydrate, Maize Starch, Hypromellose, Magnesium Stearate, Ready Coat Iron Oxide Yellow 36F0193220	2 gm	✓	✓
Lactose Monohydrate	1 gm	✓	✓
Maize starch	1 gm	✓	✓
Hypromellose	1 gm	✓	✓
Magnesium Stearate	1 gm	✓	✓
Ready Coat Yellow	1 gm	✓	✓
Purified Water	1 gm	✓	✓

1.3 FINISHED PRODUCT

1.3.1 Formula Development:

Trials conducted by using formulas which are given below:

TRIAL I:

Batch Size: 3000 Tablets (450.00 gm)

Formula:

S. No.	Composition	Specification	Qty/Tablet in mg	Qty/Batch in gm	Function
Granulation					
1.	*Solifenacin Succinate	BP	5.00	15.00	Active Substance
2.	@Maize Starch	BP	40.00	120.00	Bulking Agent
3.	Lactose Monohydrate	BP	102.50	307.50	Diluent
Binder					
4.	Hypromellose	BP	1.00	3.00	Binder
5.	#Purified water	BP	q.s.	61.66	Solvent
Lubrication					
6.	Magnesium stearate	BP	1.50	4.50	Lubricant
Coating material					
7.	**Ready Coat Iron Oxide Yellow 36F0193220	IH	3.00	11.70	Film Former
8	#Purified water	BP	q.s.	105.00	Solvent
Average wt. of Coated tablets			153.00		

Note:* Standard Quantity change as per purity.
@ Actual quantity of starch to be calculated on the basis of actual quantity of Solifenacin Succinate
Quantity to be considered as not available in final product.
** 30% extra qty. to avoid the loss during coating

Manufacturing process:

Step 1: Weighing & dispensing of raw material

1. Weighing & dispensing of raw material is carried out as per operational raw material list & Check control no, mfg. date, exp date etc.

Step 2: Sifting of Raw Material

1. Solifenacin Succinate & lactose monohydrate geometrically mixed & then mixed in polybag. 1:1 proportion 3 times.
2. Sift Admixtures of solifenacin succinate and Lactose monohydrate and maize starch trough sieve # 40
3. Sift Magnesium stearate through 60# sieve & keep aside for lubrication stage

Step 3: Dry Mixing

Load sifted step 1, step 2 & step 3 in RMG & dry mix the materials for 10 minutes at impeller slow speed.

Step 4: Binder Solution Preparation

1. Check the cleanliness of the solution preparation vessel.
2. Take required Quantity of Purified water in binder preparation vessel then add slowly hypromellose with continuous stirring to form a clear solution. Stirring to form a clear solution.

Step 5: Granulation

1. Add the binder solution slowly into RMG at impeller slow speed mix for 1 minutes followed by racking.
2. If required Quantity of purified water in RMG at impeller slow speed mix for 1 minute followed by racking.
3. Start the kneading by operating the impeller at Fast Speed for 30 seconds.
4. Unload the material.

Step 6: Drying

1. Check the integrity of the finger bag and FBD bowl before and after drying.
2. Dry the wet granules at inlet temperature 50°C-70°C (Set Point-60°C) till the outlet temperature reaches 40 (30 to 50°C), racking to be carried out whenever required.
3. After reaching outlet temperature 40 (30 to 50°C), Check the LOD on Halogen Moisture balance at 105°C. (Limit: NMT 2.0%).
4. Continue drying to bring the moisture content within limit.

Step 7: Sizing & milling

1. Dried granules size through 20# sieve and oversized granules milled through 3, 2.5, and 2 mm screen simultaneously by the Tap cone mill, 20# retention from each milling stage should be milled by subsequent screen. At last, retaintion after 2 mm screen milling observed should be pressed out through 20# sieve.

Final granules size: 20#

2. Calculate the yield of dried granules.

Step 8: Lubrication

1. Sift the magnesium stearate, through 60# s.s sieve after completion of pre lubrication, load the sifted magnesium stearate into the bin. Blender & lubrication is done for 5 min at 8 RPM speed. Check the LOD of blend.

Step 9: Compression

1. Temperature NMT 22±3°C and Humidity NMT 50% ±10of the area shall be maintained.
2. Compress the tablet at compression weight 150 mg on punch 8.0 mm, round shape plane on both sides.
3. Record the weight of compressed tablets.

Coating Solution preparation:

1. Take the purified water.

2. Add Ready Coat Iron Oxide Yellow 36F0193220 material in purified water under stirring in solution preparation vessel under vertex forming stirring & stir for 45 minutes to form uniform coating dispersion. Filter the coating solution before use by muslin cloth.

S. No	Parameters	Set	Actual
1	Inlet Temperature	60°C±15°C	70.9°C
2	Pan Speed	2-30 RPM	4 RPM
3	Pump Speed	2-8 RPM	14 RPM
4	Atomization air	0.1-0.4 kg/cm^2	0.5

Evaluation of Product:

Lubricated Blend Parameters	
Appearance	White Colour
Bulk density	0.64
Tapped density	0.81
Compressibility Index	20.98
Hausner's ratio	1.26
Tablet (Coated) parameters	
Weight	150.3-154.2
Hardness	3.00 to 3.98
Disintegration Test	1 min 50 sec
Thickness	3.20 to 3.25

Friability	0.15% w/w
Chemical parameters	
Assay	94.5%
Dissolution	95.3%

Summary

The above batch results in physical found satisfactory and chemical parameters are not found satisfactory like assay found at lower side, hence further trial was taken to improve assay.

TRIAL II:

Batch Size: 3000 Tablets (450.00 gm)

Formula:

S. No.	Composition	Specification	Qty/Tablet in mg	Qty/Batch in gm	Function
Granulation					
1.	*Solifenacin Succinate	BP	5.00	15.00	Active Substance
2.	@Maize Starch	BP	40.00	120.00	Bulking Agent
3.	Lactose Monohydrate	BP	102.50	307.50	Diluent
Binder					
4.	Hypromellose	BP	1.00	3.00	Binder
5.	#Purified water	BP	q.s.	61.66	Solvent
Lubrication					
6.	Magnesium stearate	BP	1.50	4.50	Lubricant
Coating material					
7.	**Ready Coat Iron Oxide Yellow 36F0193220	IH	3.00	11.70	Film Former
8	#Purified water	BP	q.s.	105.00	Solvent
Average wt. of Coated tablets			153.00		

Note:* Standard Quantity change as per purity.

@ Actual quantity of starch to be calculated on the basis of actual quantity of Solifenacin Succinate

Quantity to be considered as not available in final product.

** 30% extra qty. to avoid the loss during coating

Manufacturing process:

Step 1: Weighing & dispensing of raw material

1. Weighing & dispensing of raw material is carried out as per operational raw material list & Check control no, mfg. date, exp date etc.

Step 2: Sifting of Raw Material

1. Solifenacin Succinate & lactose monohydrate geometrically mixed & then mixed in polybag. 1:1 proportion 3 times.
2. Sift Admixtures of solifenacin succinate and Lactose monohydrate and maize starch trough sieve # 40
3. Sift Magnesium stearate through 60# sieve & keep aside for lubrication stage

Step 3: Dry Mixing

Load sifted step 1, step 2 & step 3 in RMG & dry mix the materials for 10 minutes at impeller slow speed.

Step 4: Binder Solution Preparation

1. Check the cleanliness of the solution preparation vessel.
2. Take required Quantity of Purified water in binder preparation vessel then add slowly hypromellose with continuous stirring to form a clear solution. Stirring to form a clear solution.

Step 5: Granulation

1. Add the binder solution slowly into RMG at impeller slow speed mix for 1 minutes followed by racking.
2. If required Quantity of purified water in RMG at impeller slow speed mix for 1 minute followed by racking.
3. Start the kneading by operating the impeller at Fast Speed for 30 seconds.
4. Unload the material.

Step 6: Drying

1. Check the integrity of the finger bag and FBD bowl before and after drying.
2. Dry the wet granules at inlet temperature 50°C-70°C (Set Point-60°C) till the outlet temperature reaches 40 (30 to 50°C), racking to be carried out whenever required.
3. After reaching outlet temperature 40 (30 to 50°C), Check the LOD on Halogen Moisture balance at 105°C. (Limit: NMT 2.0%).
4. Continue drying to bring the moisture content within limit.

Step 7: Sizing & milling

1. Dried granules size through 20# sieve and oversized granules milled through 3, 2.5, and 2 mm screen simultaneously by the Tap cone mill, 20# retention from each milling stage should be milled by subsequent screen. At last, retaintion after 2 mm screen milling observed should be pressed out through 20# sieve.

Final granules size: 20#

2. Calculate the yield of dried granules.

Step 8: Lubrication

1. Sift the magnesium stearate, through 60# s.s sieve after completion of pre lubrication, load the sifted magnesium stearate into the bin. Blender & lubrication is done for 5 min at 8 RPM speed. Check the LOD of blend.

Step 9: Compression

1. Temperature NMT 22±3°C and Humidity NMT 50% ±10of the area shall be maintained.
2. Compress the tablet at compression weight 150 mg on punch 8.0 mm, round shape plane on both sides.
3. Record the weight of compressed tablets.

Step 10:

Coating Solution preparation:

1. Take the purified water.

2. Add Ready Coat Iron Oxide Yellow 36F0193220 material in purified water under stirring in solution preparation vessel under vertex forming stirring & stir for 45 minutes to form uniform coating dispersion. Filter the coating solution before use by muslin cloth.

S. No	Parameters	Set	Actual
1	Inlet Temperature	60°C±15°C	70.9°C
2	Pan Speed	2-30 RPM	4 RPM
3	Pump Speed	2-8 RPM	14 RPM
4	Atomization air	0.1-0.4 kg/cm^2	0.5

Evaluation of Product:

Lubricated Blend Parameters	
Appearance	White Colour
Bulk density	0.63
Tapped density	0.84
Compressibility Index	20.95

Hausner's ratio	1.25
Tablet (Coated) parameters	
Weight	150.2-154.0
Hardness	3.05 to 3.89
Disintegration Test	1 min 55 sec
Thickness	3.25 to 3.30
Friability	0.18% w/w
Chemical parameters	
Assay	101.2%
Dissolution	97.1%

Summary

The above batch results in physical and chemical parameters are found satisfactory, but need to take confirmatory batch.

TRIAL III:

Batch Size: 3000 Tablets (450.00 gm)

Formula:

S. No.	Composition	Specification	Qty/Tablet in mg	Qty/Batch in gm	Function
Granulation					
1.	*Solifenacin Succinate	BP	5.00	15.00	Active Substance
2.	@Maize Starch	BP	40.00	120.00	Bulking Agent
3.	Lactose Monohydrate	BP	102.50	307.50	Diluent
Binder					
4.	Hypromellose	BP	1.00	3.00	Binder
5.	#Purified water	BP	q.s.	61.66	Solvent
Lubrication					
6.	Magnesium stearate	BP	1.50	4.50	Lubricant
Coating material					
7.	**Ready Coat Iron Oxide Yellow 36F0193220	IH	3.00	11.70	Film Former
8.	#Purified water	BP	q.s.	105.00	Solvent
Average wt. of Coated tablets			153.00		

Note:* Standard Quantity change as per purity.
@ Actual quantity of starch to be calculated on the basis of actual quantity of Solifenacin Succinatre
Quantity to be considered as not available in final product.
** 30% extra qty. to avoid the loss during coating

Manufacturing process:

Step 1: Weighing & dispensing of raw material

1. Weighing & dispensing of raw material is carried out as per operational raw material list & Check control no, mfg. date, exp date etc.

Step 2: Sifting of Raw Material

1. Solifenacin Succinate & lactose monohydrate geometrically mixed & then mixed in polybag. 1:1 proportion 3 times.
2. Sift Add mixture of solifenacin succinate and Lactose monohydrate and maize starch trough sieve # 40
3. Sift Magnesium stearate through 60# sieve & keep aside for lubrication stage.

Step 3: Dry Mixing

Load sifted step 1, step 2 & step 3 in RMG & dry mix the materials for 10 minutes at impeller slow speed.

Step 4: Binder Solution Preparation

1. Check the cleanliness of the solution preparation vessel.
2. Take required Quantity of Purified water in binder preparation vessel then add slowly hypromellose HPMC 3mpas with continuous stirring to form a clear solution. Stirring to form a clear solution.

Step 5: Granulation

1. Add the binder solution slowly into RMG at impeller slow speed mix for 1 minutes followed by racking.
2. If required Quantity of purified water in RMG at impeller slow speed mix for 1 minute followed by racking.
3. Start the kneading by operating the impeller at Fast Speed for 30 seconds.
4. Unload the material.

Step 6: Drying

1. Check the integrity of the finger bag and FBD bowl before and after drying.
2. Dry the wet granules at inlet temperature 50°C-70°C (Set Point-60°C) till the outlet temperature reaches 40(30 to 50°C), racking to be carried out whenever required.
3. After reaching outlet temperature 40 (30 to 50°C), Check the LOD on Halogen Moisture balance at 105°C. (Limit: NMT 2.0%).

4. Continue drying to bring the moisture content within limit.

Step 7: Sizing & milling

1. Dried granules size through 20# sieve and oversized granules milled through 3, 2.5, and 2 mm screen simultaneously by the Tap cone mill, 20# retention from each milling stage should be milled by subsequent screen. At last, retaintion after 2 mm screen milling observed should be pressed out through 20# sieve.

Final granules size: 20#

2. Calculate the yield of dried granules.

Step 8: Lubrication

1. Sift the magnesium stearate, through 60# s.s sieve after completion of pre lubrication, load the sifted magnesium stearate into the bin. Blender & lubrication is done for 5min at 8RPM speed. Check the LOD of blend.

Step 9: Compression

2. Temperature NMT 22±3°C and Humidity NMT 50% ±10of the area shall be maintained.
3. Compress the tablet at compression weight 150 mg on punch 8.0 mm, round shape plane on both sides.
4. Record the weight of compressed tablets.

Step 10:

Coating Solution preparation:

1. Take the purified water.

2. Add Ready Coat Iron Oxide Yellow 36F0193220 material in purified water under stirring in solution preparation vessel under vertex forming stirring & stir for 45 minutes to form uniform coating dispersion. Filter the coating solution before use by muslin Cloth

S. No	Parameters	Set	Actual
1	Inlet Temperature	60°C±15°C	70.9°C
2	Pan Speed	2-30 RPM	4 RPM
3	Pump Speed	2-8 RPM	14 RPM
4	Atomization air	0.1-0.4 kg/cm^2	0.5

LEBEN LIFE SCIENCES	Leben Life Sciences Pvt. Ltd. Akola – 444104 (MS) INDIA
Title	PRODUCT DEVELOPMENT REPORT
Product	SOLIFENACIN SUCCINATE TABLETS 5 MG

Evaluation of Product:

Lubricated Blend Parameters	
Appearance	White Colour
Bulk density	0.65
Tapped density	0.82
Compressibility Index	20.73
Hausner's ratio	1.35
Tablet (Coated) Parameters	
Hardness	3.10 to 3.98
Thickness	3.16 to 3.19
Friability	0.06%

Sr. No.	Test	Specification	Observation
1.	**Description**	Light yellow coloured, round shaped, film coated tablet, plain on both sides.	Light yellow coloured, round shaped, film coated tablet, plain on both sides.
2.	**Identification By IR**	The infrared absorption spectrum of the dried residue is concordant with the reference spectrum of Solifenacin Succinate RS/WS.	Complies
3.	**Average Weight**	153.00 mg ± 7.5% (141.53 mg to 164.48 mg)	153.97 mg
4.	**Content Uniformity**	AV NMT 15	1.55%
5.	**Disintegration Test**	Not more than 30 minutes	1 Min 30 Sec
6.	**Dissolution**	NLT 75% (Q)	95.4%
7.	**Assay**	95.0% to 105.0% of the stated amount	99.2%
8.	**Microbial Limit Test**	TAMC: NMT 1000 cfu/gm. TYMC: NMT 100 cfu/gm. Escherichia coli: absent/gm. Salmonella: absent/gm. P. aeruginosa: absent/gm. S. aureus : absent/gm.	10 cfu/gm < 20 cfu/gm absent/gm absent/gm absent/gm absent/gm

Summary

The above batch results in physical and chemical parameters were found to be satisfactory.

LEBEN LIFE SCIENCES	**Leben Life Sciences Pvt. Ltd.** Akola – 444104 (MS) INDIA
Title	**PRODUCT DEVELOPMENT REPORT**
Product	**SOLIFENACIN SUCCINATE TABLETS 5 MG**

PROTOTYPE FORMULA

S. No.	Composition	Specification	Qty/Tablet in mg	Qty/Batch in gm	Function
Granulation					
1.	*Solifenacin Succinate	BP	5.00	15.00	Active Substance
2.	@Maize Starch	BP	40.00	120.00	Bulking Agent
3.	Lactose Monohydrate	BP	102.50	307.50	Diluent
Binder					
4.	Hypromellose	BP	1.00	3.00	Binder
5.	#Purified water	BP	q.s.	61.66	Solvent
Lubrication					
6.	Magnesium stearate	BP	1.50	4.50	Lubricant
Coating material					
7.	**Ready Coat Iron Oxide Yellow 36F0193220	IH	3.00	11.70	Film Former
8.	#Purified water	BP	q.s.	105.00	Solvent
Average wt. of Coated tablets			153.00		

Note:* Standard Quantity change as per purity.
@ Actual quantity of starch to be calculated on the basis of actual quantity of Solifenacin Succinate
Quantity to be considered as not available in final product.
** 30% extra qty. to avoid the loss during coating

Formula finalization:

Trials were taken with varying quantities of material to optimize the formulation. All the III trials were tested for physical parameters & coating parameters studied for dissolution profile. Among the selected lab trials, trial no. III was found to meet targeted product profile such as assay, dissolution test etc. Hence trial no.III was finalized as final formulation.

Conclusion:

Based on results of Preformulation study and developmental trials taken, trials no. III was selected as final formulation. Further process optimization studies have also been conducted on this trial no. III to finalize the manufacturing process.

1.3.2 Overages

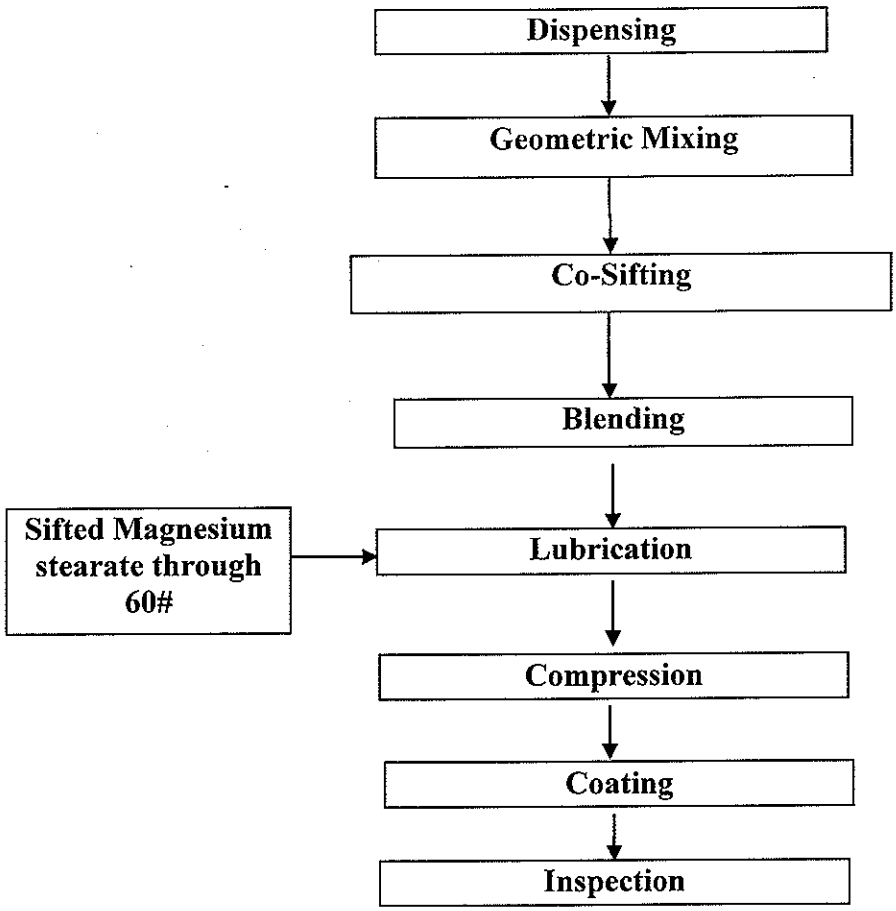
No overages added in the formulation.

8.3.3 Physiochemical and Biological Properties

Sr. No.	Tests	Specifications
1.	**Description**	Light yellow coloured, round shaped, film coated tablet, plain on both sides.
2.	**Identification By IR**	The infrared absorption spectrum of the dried residue is concordant with the reference spectrum of Solifenacin Succinate RS/WS.
3.	**Average weight**	153.00 mg ± 7.5% (141.53 mg to 164.48 mg)
4.	**Uniformity of weight**	Not More than 2 tablets deviate from average weight by 7.5% None Tablet deviate from the average weight by more than 15%
5.	**Content Uniformity**	AV NMT 15
6.	**Hardness**	NLT 3 Kp
7.	**Friability**	NMT 1.0% w/w
8.	**Disintegration time**	NMT 30 minutes
9.	**Dissolution**	NLT 75% (Q) of the labeled amount
10.	**Assay**	95.0% to 105.0% of the stated amount
11.	**Microbial Limit Test**	TAMC: NMT 1000 cfu/gm. TYMC: NMT 100 cfu/gm. Escherichia coli: absent/gm. Salmonella: absent/10 gm. P. aeruginosa: absent/gm. S. aureus : absent/gm.

1.4 MANUFACTURING PROCESS DEVELOPMENT

Manufacturing Process flow chart:



STANDARD MANUFACTURING PROCEDURE INSTRUCTIONS:

Before starting Process ensure the following:

1. Check &ensure removal of previous product.
2. Check & ensure area cleaning.
3. Check & ensure wall & roof are clean.
4. Check & ensure all utensils, machines and equipment's are clean & fit for use.
5. Check & ensure balances are calibrated.
6. Temperature - 22°C (±3°C).
7. Relative Humidity 50%(±10%).
8. Check &ensure operator has worn nose mask, caps & gloves.

MANUFACTURING PROCESS:

Before manufacturing operation is begun:

01. All machineries and equipment used for manufacturing process are cleaned as per respective standard operating procedure for cleaning.
02. The manufacturing area to be used is cleaned and made suitable for operations.
03. All the process, products name are properly displayed in notified area.
04. Safety gadgets like hand gloves, mask and slippers provided to workmen.

Manufacturing Steps:

Standard Manufacturing process:

Step 1: Weighing & dispensing of raw material

1. Weighing & dispensing of raw material is carried out as per operational raw material list & Change control no, mfg. date, exp date etc.

Step 2: Sifting of Raw Material

1. Solifenacin Succinate & lactose monohydrate geometrically mixed & then mixed in polybag. 1:1 proportion 3 times.
2. Sift Addmixture of solifenacin succinate and Lactose monohydrate and maize starch trough sieve # 40
3. Sift Magnesium stearate through 60# sieve & keep aside for lubrication stage.

Step 3: Dry Mixing

1. Load sifted step 1, step 2 & step 3 in RMG & dry mix the materials for 10 minutes at impeller slow speed.

Step 4: Binder Solution Preparation

1. Check the cleanliness of the solution preparation vessel.

2. Take required Quantity of Purified water in binder preparation vessel then add slowly hypromellose HPMC 3mpas with continuous stirring to form a clear solution. Stirring to form a clear solution.

Step 5: Granulation

1. Add the binder solution slowly into RMG at impeller slow speed mix for 1 minutes followed by racking.
2. If required Quantity of purified water in RMG at impeller slow speed mix for 1 minute followed by racking.
3. Start the kneading by operating the impeller at Fast Speed for 30 seconds.
4. Unload the material.

Step 6: Drying

1. Check the integrity of the finger bag and FBD bowl before and after drying.
2. Dry the wet granules at inlet temperature 50°C-70°C (Set Point-60°C) till the outlet temperature reaches 40(30 to 50°C), racking to be carried out whenever required.
3. After reaching outlet temperature 40 (30 to 50°C), Check the LOD on Halogen Moisture balance at 105°C. (Limit: NMT 2.0%).
4. Continue drying to bring the moisture content within limit.

Step 7: Sizing & milling

1. Dried granules size through 20# sieve and oversized granules milled through 3, 2.5, and 2 mm screen simultaneously by the Tap cone mill, 20# retention from each milling stage should be milled by subsequent screen. At last, retaintion after 2 mm screen milling observed should be pressed out through 20# sieve.

Final granules size: 20#

2. Calculate the yield of dried granules.

Step 8: Lubrication

1. Sift the magnesium stearate, through 60# s.s sieve after completion of pre lubrication, load the sifted magnesium stearate into the bin. Blender & lubrication is done for 5min at 8RPM speed. Check the LOD of blend.

Step 9: Compression

1. Temperature NMT 22±3°C and Humidity NMT 50% ±10of the area shall be maintained.
2. Compress the tablet at compression weight 150 mg on punch 8.0 mm, round shape plane on both sides.
3. Record the weight of compressed tablets.

Step 10:

Coating Solution preparation:

1. Take the purified water.
2. Add Ready Coat Iron Oxide Yellow 36F0193220 material in purified water under stirring in solution preparation vessel under vertex forming stirring & stir for 45 minutes to form uniform coating dispersion. Filter the coating solution before use by muslin Cloth

S. No	Parameters	Set	Actual
1	Inlet Temperature	60°C±15°C	70.9°C
2	Pan Speed	2-30 RPM	4 RPM
3	Pump Speed	2-8 RPM	14 RPM
4	Atomization air	0.1-0.4 kg/cm^2	0.5

3. Sent the sample to ADL for analysis.

Step 11: Packing

Receiving release note from ADL and pack the ointment as per the proposed packing schedule.

PARAMETER	CONDITION
Room Temperature	NMT 23°C ± 3°C
Relative Humidity	NMT 50 %RH ± 10%%RH

1.5 CONTAINER CLOSURE SYSTEM

Evaluation of suitable Container Closure System

The proposed pack for trial batches of Solifenacin Succinate Tablets 5 mg is ALU/ALU and ALU/PVC are packed in a primary carton.

These packing materials, which are in common use for packaging of pharmaceutical Tablets preparations, were chosen with reference to commercial presentation required so as to provide better protection against moisture vapors penetration and also as to achieve the required performance characteristics of the product use.

Protection:

Aluminum & PVC foil gave satisfactory protection against temperature and moisture to the tablet.

Compatibility:

The Aluminum & PVC foil met Pharmacopoeia requirements indicating compatibility with tablet.

Safety:

There are no safety issues with proposed commercial packing material. They are listed as generally recognized as safe (GRAS)" or "suitable for direct or indirect contact with food."

1.6 MICROBIOLOGICAL ATTRIBUTES

Solifenacin Succinate Tablets 5 mg Tablets is routinely tested in accordance with Relevant Pharmacopoeial requirements for pharmaceutical preparation for oral use, the specification limit for which are as follow:

Parameters	Specifications
Total Aerobic Microbial Count (TAMC)	1000 cfu/gm
Total Yeasts and Molds Count (TYMC)	100 cfu/gm
Escherichia coli	Absent/gm
Salmonella	Absent/10 gm
P. aeruginosa	Absent/gm
S. aureus	Absent/gm

Manufacturing of product is carried under appropriate as conditions, which are routinely confirmed/controlled by suitable environmental controls and monitoring activities.

1.7 COMPATIBILITY

The compatibility of the drug product with reconstitution diluents should be addressed to provide appropriate and supportive information for the labeling.

Not Applicable.

RESULT OF EVALUATION OF FORMULA

A strategy for the development of Solifenacin Succinate Tablets 5 mg Tablets was devised based on the information from literature and RLD formulation considerations. The formula and process was refined as result of experience gained during development trials.

The formula and process develop in adequate to produce the formulations meeting all the quality specifications.

LEBEN LIFE SCIENCES	**Leben Life Sciences Pvt. Ltd.** Akola – 444104 (MS) INDIA
Title	**PRODUCT DEVELOPMENT REPORT**
Product	**SOLIFENACIN SUCCINATE TABLETS 5 MG**

REPORT APPROVAL

Activity	Name	Department	Designation	Signature & Date
Prepared By	Kishor Rakade	FDL	Trainee	Rakade 22/02/2021
Checked By	Prashant Gurnule	FDL	Officer	22/02/2021
Reviewed By	Syed Iftekhar	ADL	Executive	22/02/2021
Verified By	Naved Ahmed	RA	Executive	22/02/2021